SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40 **Total Marks:40**

Annexure A

Mock Interview

QUESTIONS:40 **Total Marks:40**

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)

27. How would you present your simulation results to a client? (BL4)(1)
28. Explain jitter using a real-world analogy (BL5)(1)
29. Describe how you tested network performance in a lab (BL4)(1)
30. Explain the importance of performance metrics in one structured answer (BL5)(1)
31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
32. Why are you interested in networking and simulation tools? (BL5)(1)
33. What makes you confident in analyzing network performance? (BL5)(1)
34. Explain a concept you are very strong in (related to networking) (BL5)(1)
35. How would you handle a question you don't know? (BL5)(1)
36. Describe a situation where you solved a technical problem (BL4)(1)
37. Why should we select you for a networking role? (BL5)(1)
38. What challenges did you face while learning Packet Tracer? (BL5)(1)
39. How do you improve your technical skills regularly? (BL5)(1)
40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	**30%**	**Demonstrates thorough understanding; answers all questions accurately**	**Shows good understanding with few errors**	**Partial understanding; several incorrect responses**	**Lacks understanding; majority of answers incorrect**
Accuracy of Answers	**25%**	**All answers are correct**	**Most answers are correct with minor mistakes**	**Some correct answers; frequent errors**	**Mostly incorrect answers**
Application of Knowledge	**20%**	**Correctly applies concepts to problem-based questions**	**Applies concepts with minor errors**	**Limited ability to apply concepts**	**Unable to apply concepts**
Time Management	**15%**	**Completes quiz well within time efficiently**	**Completes on time with minor delays**	**Requires extra time or rushes through questions**	**Unable to complete within given time**
Question Interpretation	**10%**	**Interprets all questions correctly and responds appropriately**	**Misinterprets a few questions**	**Often misinterprets questions**	**Unable to understand questions**

Answers

1. Network congestion increases packet delay because packets wait in queues. When buffers become full, packets are dropped, causing packet loss.
2. **Star:** Easy to manage and reliable, but failure of the central switch affects the network. **Mesh:** Provides multiple paths, so it has higher reliability and fault tolerance.
3. Each router adds processing, queuing, and transmission delay. Therefore, more routers increase the total end-to-end delay.
4. QoS prioritizes voice and video packets, reducing delay, jitter, and packet loss. This provides smoother calls and video streaming.
5. Fiber optic cables provide higher speed, greater bandwidth, longer-distance transmission, and better resistance to electromagnetic interference than copper cables.
6. VLANs divide a network into logical segments. This reduces unnecessary broadcast traffic and improves security by isolating different groups.
7. **TCP** is reliable and connection-oriented but has more overhead. **UDP** is faster with less overhead but does not guarantee delivery, making it suitable for real-time applications.
8. Load balancing distributes traffic across multiple devices or links. It prevents overload, improves performance, and increases network availability.
9. Frequent routing updates consume bandwidth and router processing resources. Excessive updates can reduce network efficiency.
10. Firewalls and encryption improve security but require additional processing, which can slightly reduce network performance. Proper configuration provides a balance between security and speed.
11. I would use **ping, traceroute, and network monitoring tools** to locate the source of delay. Then I would check congestion, routing, bandwidth, and faulty devices and fix the identified issue.
12. Possible reasons include congestion, packet loss, high latency, inefficient protocols, hardware limitations, interference, or server-side limitations.
13. I would reduce unnecessary traffic, increase bandwidth, apply QoS, optimize routing, remove bottlenecks, and monitor overloaded network devices.
14. I would check packet loss, jitter, latency, bandwidth usage, connectivity, and QoS settings. Then I would identify and fix the congested or faulty link/device.
15. I would prioritize real-time traffic using QoS, reduce congestion, increase bandwidth, optimize routing, and remove unstable network links.
16. In Packet Tracer, I would create a network, generate traffic between devices, increase the traffic load, and observe packet delay, packet loss, and congestion effects.
17. I would check the router's CPU and memory usage, interface status, traffic load, routing table, queue length, and packet statistics. Then I would remove the identified bottleneck.
18. I would use a stable high-bandwidth connection, minimize latency and jitter, reduce packet loss, use QoS to prioritize gaming traffic, and avoid network congestion.
19. I would create PCs, switches, and a router in Packet Tracer. Then I would measure **latency, throughput, packet loss, jitter, and bandwidth utilization** using suitable tests and simulation results.
20. I would use ping and traceroute to identify where the delay occurs. I would then check congestion, routing paths, bandwidth, faulty devices, and long-distance links.
21. Network performance means how well a network delivers data. A good network transfers data quickly, reliably, and with minimum delay or loss.
22. Throughput is the actual amount of data transferred per second. For example, a water pipe may have a large capacity, but the actual water flowing through it represents throughput.
23. Packet loss causes missing or corrupted parts of a video. This can result in buffering, freezing, reduced quality, and interruptions.
24. The main delay types are **processing delay, queuing delay, transmission delay, and propagation delay**.
25. Cisco Packet Tracer is a network simulation tool used to design, configure, test, and troubleshoot network topologies without requiring physical networking equipment.
26. **Simply:** QoS gives important traffic higher priority. **Technically:** QoS classifies and prioritizes traffic to control bandwidth, delay, jitter, and packet loss.
27. I would present the topology, testing method, performance metrics, graphs or tables, key findings, problems identified, and recommended improvements.
28. Jitter is the variation in packet arrival time. Like a bus that should arrive every 10 minutes but sometimes arrives after 5 minutes and sometimes after 20 minutes.

29. I created the network topology in the lab, configured the devices, generated traffic, tested connectivity, and measured parameters such as delay, throughput, packet loss, and jitter.
30. Performance metrics help evaluate whether a network is fast, reliable, and efficient. Important metrics include **bandwidth, throughput, delay, jitter, packet loss, and latency**.
31. I am a computer engineering student with knowledge of computer networks and network performance. I understand concepts such as latency, throughput, packet loss, jitter, QoS, and basic network simulation using Packet Tracer.
32. I am interested in networking because it helps devices communicate efficiently. Simulation tools like Packet Tracer allow me to design, test, and troubleshoot networks safely.
33. I am confident because I understand key performance metrics and can use logical troubleshooting methods and simulation tools to identify network problems.
34. I am strong in understanding **network performance metrics**. I can explain and analyze bandwidth, throughput, delay, jitter, and packet loss and relate them to actual network performance.
35. I would honestly say that I do not know the answer yet. I would explain what I know about the related concept and then learn the correct answer.
36. I faced a technical problem while configuring a network in Packet Tracer. I checked the device configuration and connectivity step by step, identified the incorrect setting, corrected it, and verified the connection.
37. You should select me because I have a good understanding of networking concepts, I approach technical problems logically, and I am willing to learn and improve continuously.
38. The main challenges were understanding device configuration, routing, troubleshooting connectivity, and interpreting simulation results. Practice helped me overcome these difficulties.
39. I regularly improve my technical skills through practical exercises, coding, networking simulations, online learning, and by practicing troubleshooting problems.
40. I explain complex concepts by breaking them into simple steps and using real-life examples. For example, I explain **jitter as irregular packet arrival time**, similar to a bus arriving at unpredictable intervals.